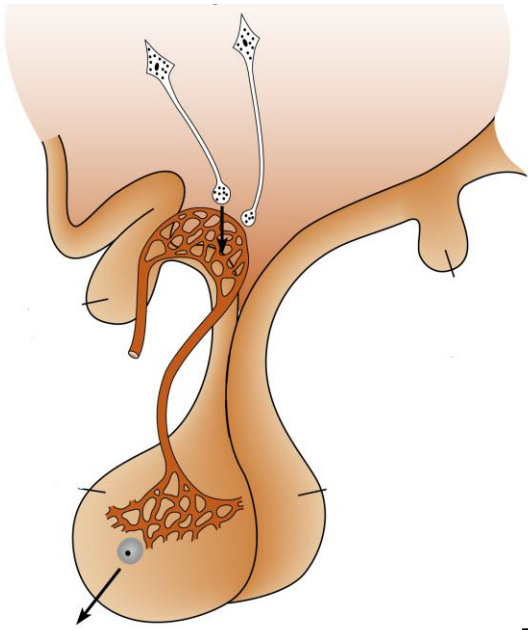
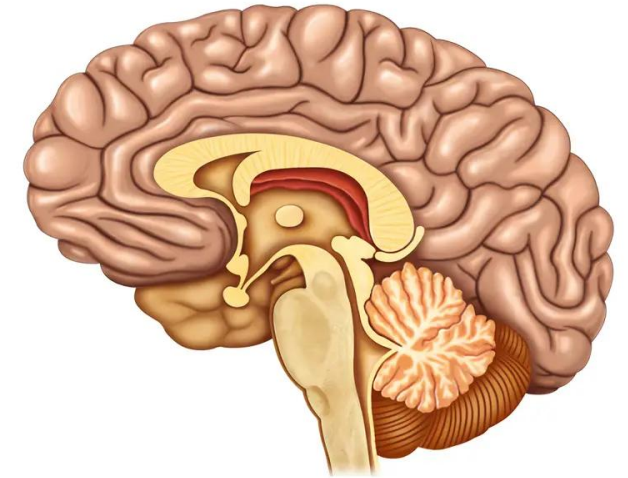




BPM 2

ER: Lecture 03



ANTERIOR & POSTERIOR PITUITARY FUNCTIONS

Subramanya Upadhy,
Professor

Depart of Physiology, Neuroscience & Behavioral Sciences
School of Medicine, St. George's University

Email: supadhy@sgu.edu

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Recommended Reading

Chapter 31: Hypothalamus and Pituitary gland. Medical Physiology: Principles for Clinical Medicine, 6e. Rodney A. Rhoades, David R. Bell.

Slides with blue background are the supplemental information of the previous slides and thus, might not be explained again during the lecture time. However, please read the slides for clarification as they are equally important to know

Glossary

ACTH	Adrenocorticotrophic hormone (corticotropin)
ADH	Antidiuretic hormone
AVP	Arginine-Vasopressin
SON	Supraoptic neuron
PVN	Paraventricular neuron
CRH	Corticotropin-releasing hormone
FSH	Follicle-stimulating hormone
GABA	Gamma-aminobutyric acid
GH	Growth hormone (somatotropin)
GHBP	Growth hormone–binding protein
GHRH	Growth hormone–releasing hormone
SRIF	Somatotropin release inhibiting factor (Somatostatin)
GnRH	Gonadotropin-releasing hormone

hCG	Human chorionic gonadotropin
hPL	Human placental lactogen
IGF	Insulin-like growth factor
LH	Luteinizing hormone
MSH	Melanocyte-stimulating hormone
PIH	Prolactin-inhibiting hormone (dopamine)
PRH	Prolactin-releasing hormone (TRH)
POMC	Pro-opiomelanocortin
PRL	Prolactin
SHBG	Sex hormone–binding globulin
SIADH	Syndrome of inappropriate secretion of antidiuretic hormone
TRH	Thyrotropin-releasing hormone
TSH	Thyroid-stimulating hormone (thyrotropin)

Learning Objectives

SOM.MK.II.BPM2.1.ER.1.PHYS.1014. Describe the effects of loss of all pituitary hormones and effects of injury to pituitary stalk.

SOM.MK.II.BPM2.1.ER.1.PHYS.1015. Describe the chemical nature, synthesis and release of prolactin.

SOM.MK.II.BPM2.1.ER.1.PHYS.1016. Explain the mechanism of action of prolactin at its target cells and list its physiological functions on 2 key target tissues.

SOM.MK.II.BPM2.1.ER.1.PHYS.1017. Describe symptoms of prolactinomas and prolactin imbalance.

SOM.MK.II.BPM2.1.ER.1.PHYS.1018. List 2 posterior pituitary hormones and discuss their structural similarity.

Learning Objectives

SOM.MK.II.BPM2.1.ER.1.PHYS.1019. Explain the regulation of vasopressin secretion.

SOM.MK.II.BPM2.1.ER.1.PHYS.1020. List the 2 target cells for vasopressin and explain why vasopressin is also known as antidiuretic hormone.

SOM.MK.II.BPM2.1.ER.1.PHYS.1021. Identify disease states caused by a) over secretion, and b) under secretion of vasopressin and list the principal symptoms of each.

SOM.MK.II.BPM2.1.ER.1.PHYS.1022. Name the stimuli for oxytocin release in relation to its reproductive and lactation functions.

SOM.MK.II.BPM2.1.ER.1.PHYS.1023. List the 2 target organs and the functional effects of oxytocin.

Lecture Outline

- ❖ **Loss of anterior pituitary functions**
- ❖ **All about prolactin**
- ❖ **Posterior pituitary hormone: Vasopressin**
- ❖ **Posterior pituitary hormone: Oxytocin**

Loss of Pituitary Function: Hypopituitarism

- Hypopituitarism means deficiency of one or more hormones of the anterior pituitary.
- **Panhypopituitarism** means a condition of **inadequate** or **absent** production of **all** anterior pituitary hormones.
- May be congenital or acquired (occur later in life) defects of pituitary gland.
- May be due to impaired functioning of the hypothalamus to secrete hypophysiotropic hormones.
- **Most common** causes: Gene mutation, tumors, blockage in blood flow, and chronic inflammation.
- Clinical features: depend on what hormone/s is/are deficient.

Loss of Pituitary Function: Clinical Examples

Sheehan syndrome:

Ischemic necrosis of the anterior pituitary following severe blood loss during labor. There may be panhypopituitarism or partial hypopituitarism. Woman presents with failure to lactate, amenorrhea, cold intolerance, fatigue & anemia. Posterior pituitary hormones may be normal. Prolactin deficiency results in inability to breastfeed.

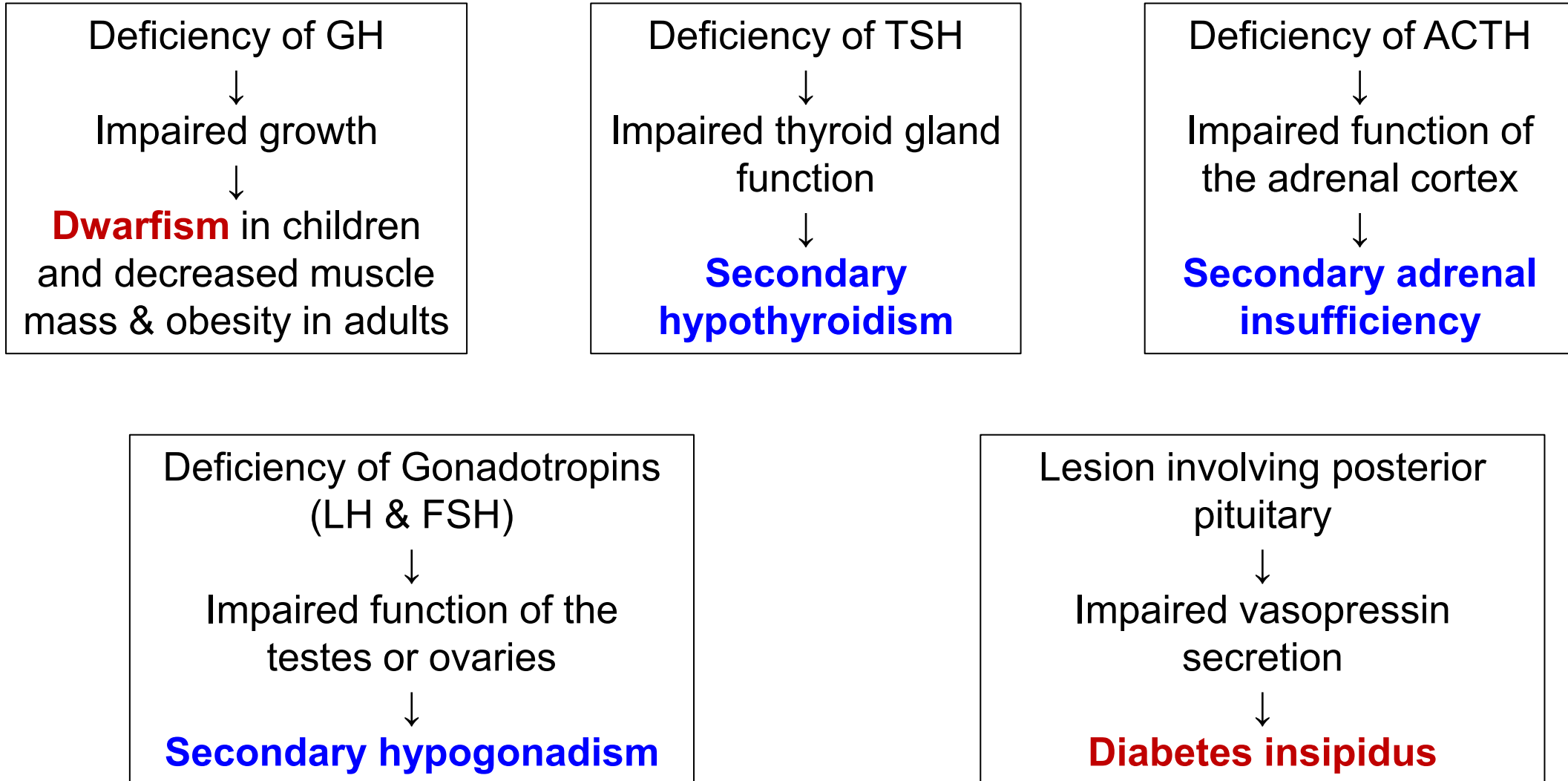
Pituitary apoplexy:

Sudden hemorrhage of pituitary gland, often in the presence of an existing pituitary tumor. Patient presents with sudden onset severe headache, visual impairment, double vision (due to III CN injury) and features of hypopituitarism. Sudden deficiency of ACTH may cause severe hypotension. MRI shows hemorrhage in sella turcica.

Empty sella syndrome:

An enlarged sella turcica that is not entirely filled with pituitary tissue. Atrophy or loss of pituitary with no known cause or secondary to surgery, radiation or tumor. Condition may be associated with elevated intracranial pressure.

Loss of Pituitary Function: Hypopituitarism



GH deficiency in children: Dwarfism

- Impaired GH secretion by the anterior pituitary (**pituitary dwarf**). These children show decreased rate of growth and short stature, but normal brain development and intellectuality. Patients have **decreased** plasma GH and IGF-1.
- Defective GH receptors in the target tissue. This is called **Laron dwarfism**, a type of GH insensitivity syndrome. Physiologic effects of GH are lost, including secretion of IGF-1 from the liver. Patients have **decreased** plasma IGF-1 but **elevated** plasma GH.



7-year-old girl patient (left) with her younger sister, 6-year-old normal (right)

GH deficiency adults

- If the GH deficiency occurs during childhood, the patient continues to have short stature and will have *miniature built* with normal intellectuality (**pituitary dwarf**).
- If the GH deficiency occurs after the fusion of epiphyses of long bones (after the age of about 20), *height is not affected*. These patients develop increased body fat, reduced muscle mass, fatigue and decreased physical fitness.



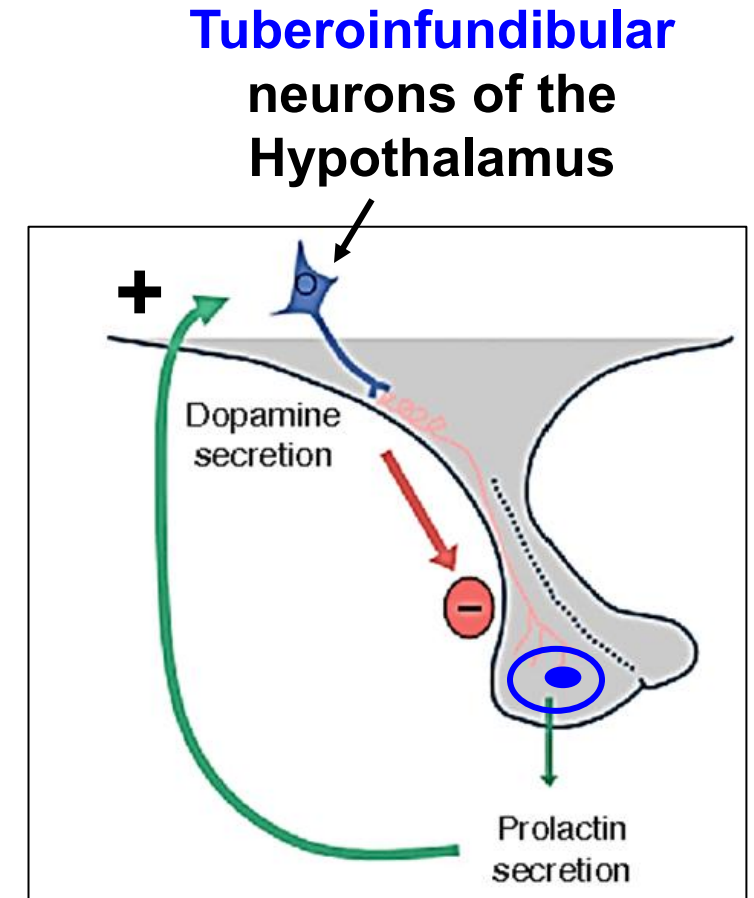
This adult man who had GH deficiency in childhood has miniature built but normal brain development

Lecture Outline

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- ❖ **All about prolactin**
- ❖ **Posterior pituitary hormone: Vasopressin**
- ❖ **Posterior pituitary hormone: Oxytocin**

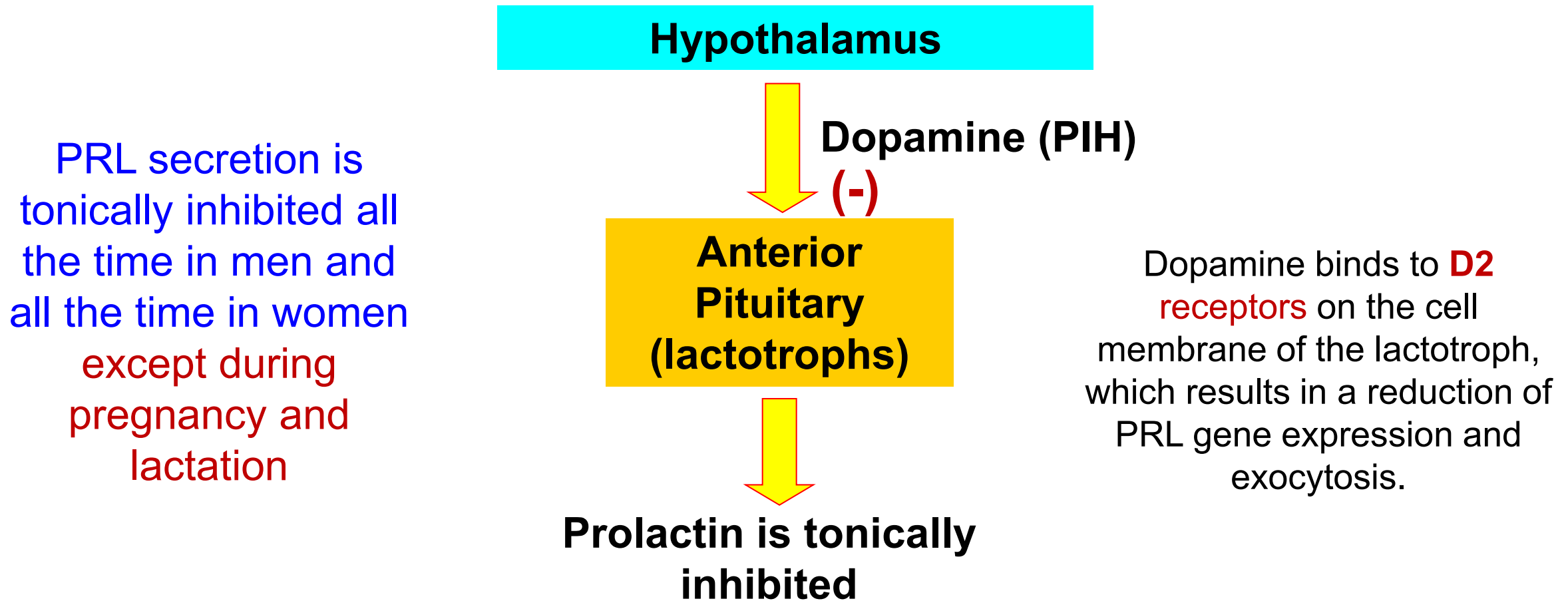
Prolactin (PRL): Synthesis, release and chemical nature

- PRL is a peptide hormone secreted by the lactotrophs.
- Belongs to GH family, structurally resembles human GH but does not have growth-promoting effects.
- PRL is **predominantly inhibited by hypothalamic Dopamine** (also known as prolactin inhibiting hormone, PIH) in both males and females.
- Dopamine inhibits PRL secretion through **D2** receptors on the lactotrophs.
- Normal blood level is very low in both sex.
- PRL level increases only during **pregnancy and breast feeding** in women.



Prolactin **inhibits** its own secretion by stimulating dopamine from the hypothalamus

Regulation of prolactin under physiologic conditions

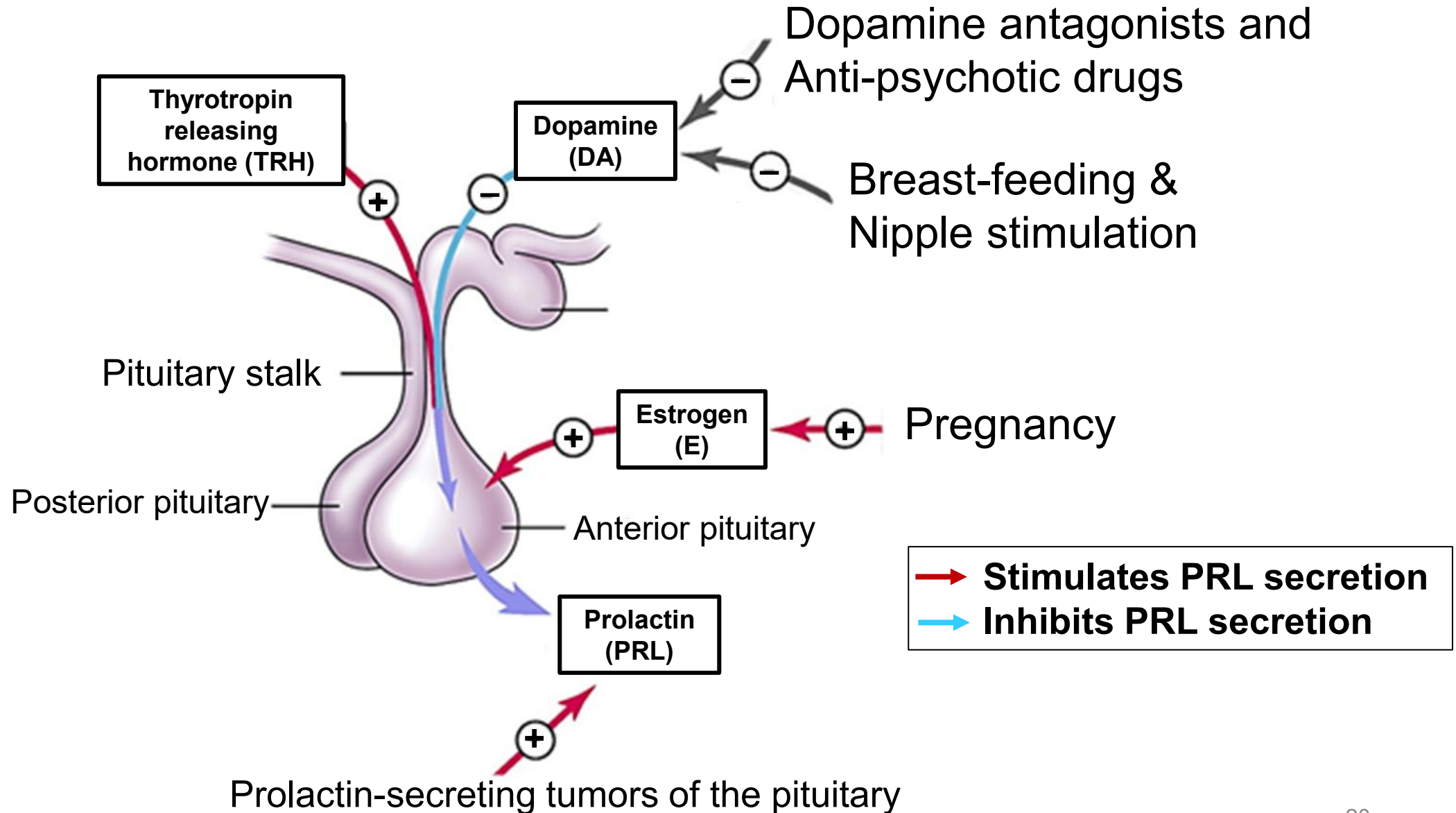


Actions of Prolactin on target tissue

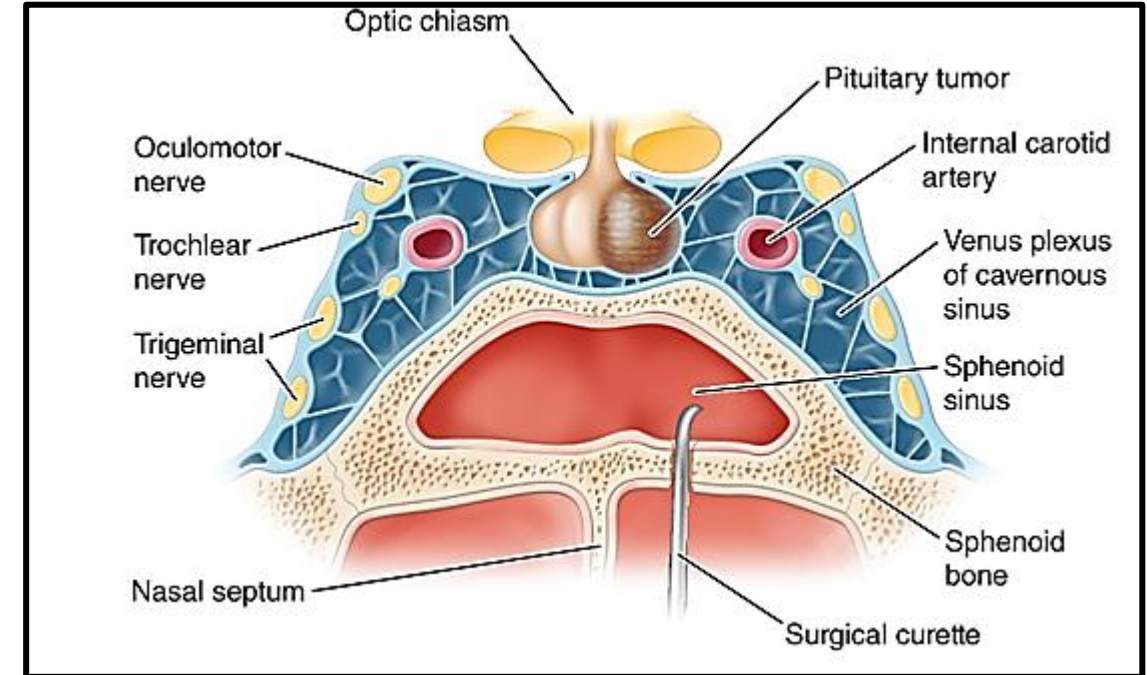
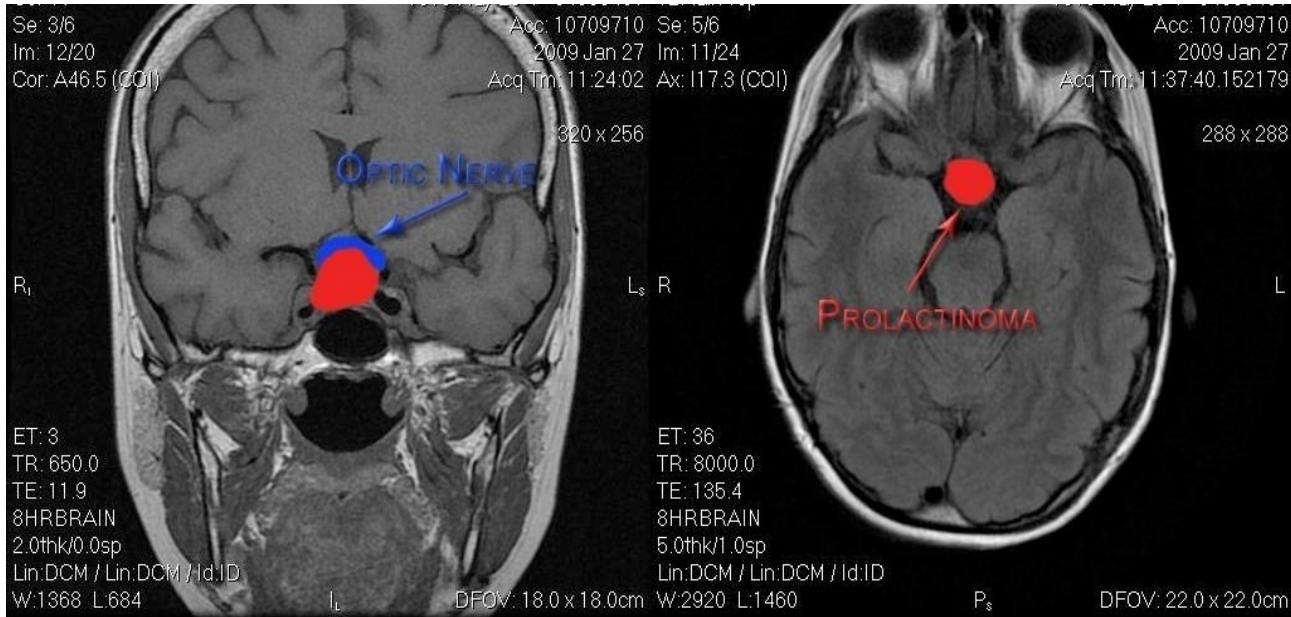
- Main function is **breast development and milk production (lactogenesis)**, **but NOT milk ejection**.
- It initiates and maintains milk production (lactation) following childbirth.
- Inhibits reproductive function in both sex by suppressing GnRH in the hypothalamus.
- PRL suppresses sexual drive in both sex.
- In the ovary, PRL causes anovulation.

Clinical correlate: High serum prolactin produces **infertility** in both men and women.
Women: Low sex drive and temporary cessation of menstruation (amenorrhea);
Men: Low testosterone, low sperm count and decreased libido.

Factors Influencing Prolactin secretion



Prolactinomas: Most common pituitary tumor in adults



In Male:

- Loss of libido,
- Low sperm count,
- Erectile dysfunction,
- Low testosterone,
- Gynecomastia.

In Female:

- Loss of menstruation,
- Anovulation,
- Loss of sexual drive,
- Galactorrhea.

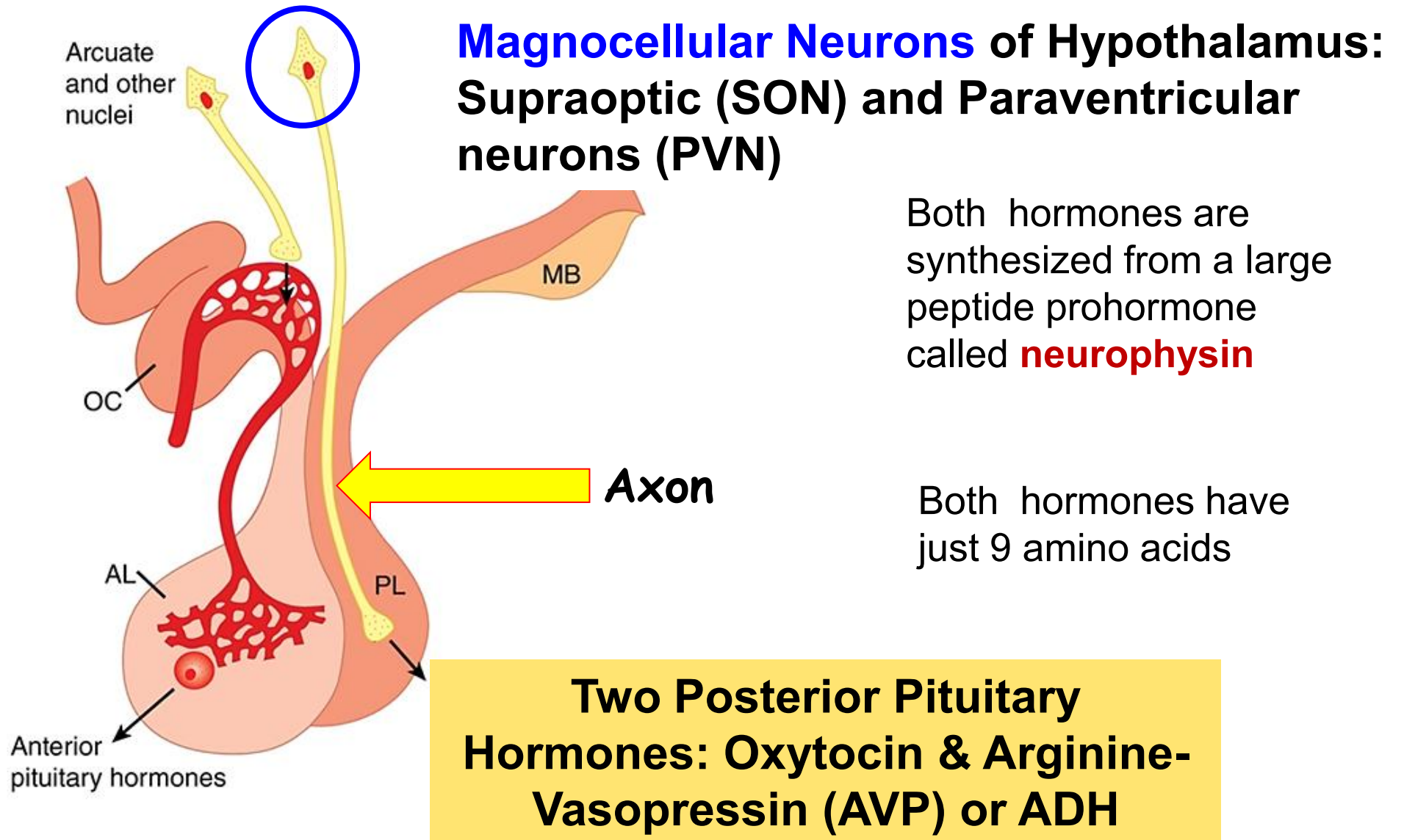
In both sex:

- Visual defects,
- Headache,
- Infertility and
- Secondary hypogonadism

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- ❖ Posterior pituitary hormone: Oxytocin

Posterior pituitary hormones



Factors regulating AVP (ADH) secretion

Stimulates ADH Secretion

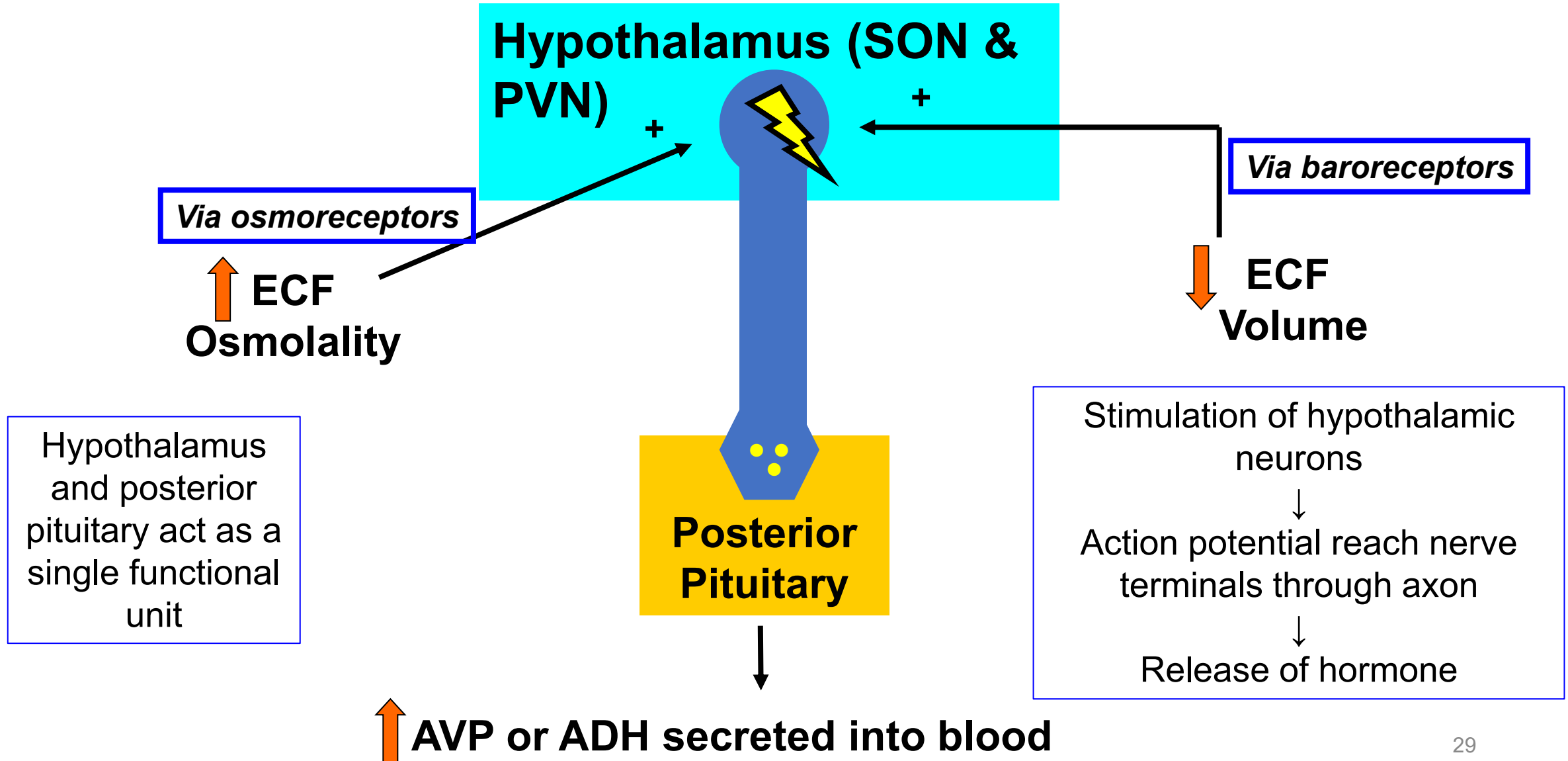
Increased body fluid osmolality
Decreased blood volume
Decreased blood pressure
Angiotensin II
Pain
Stress
Nausea and vomiting

Inhibits ADH Secretion

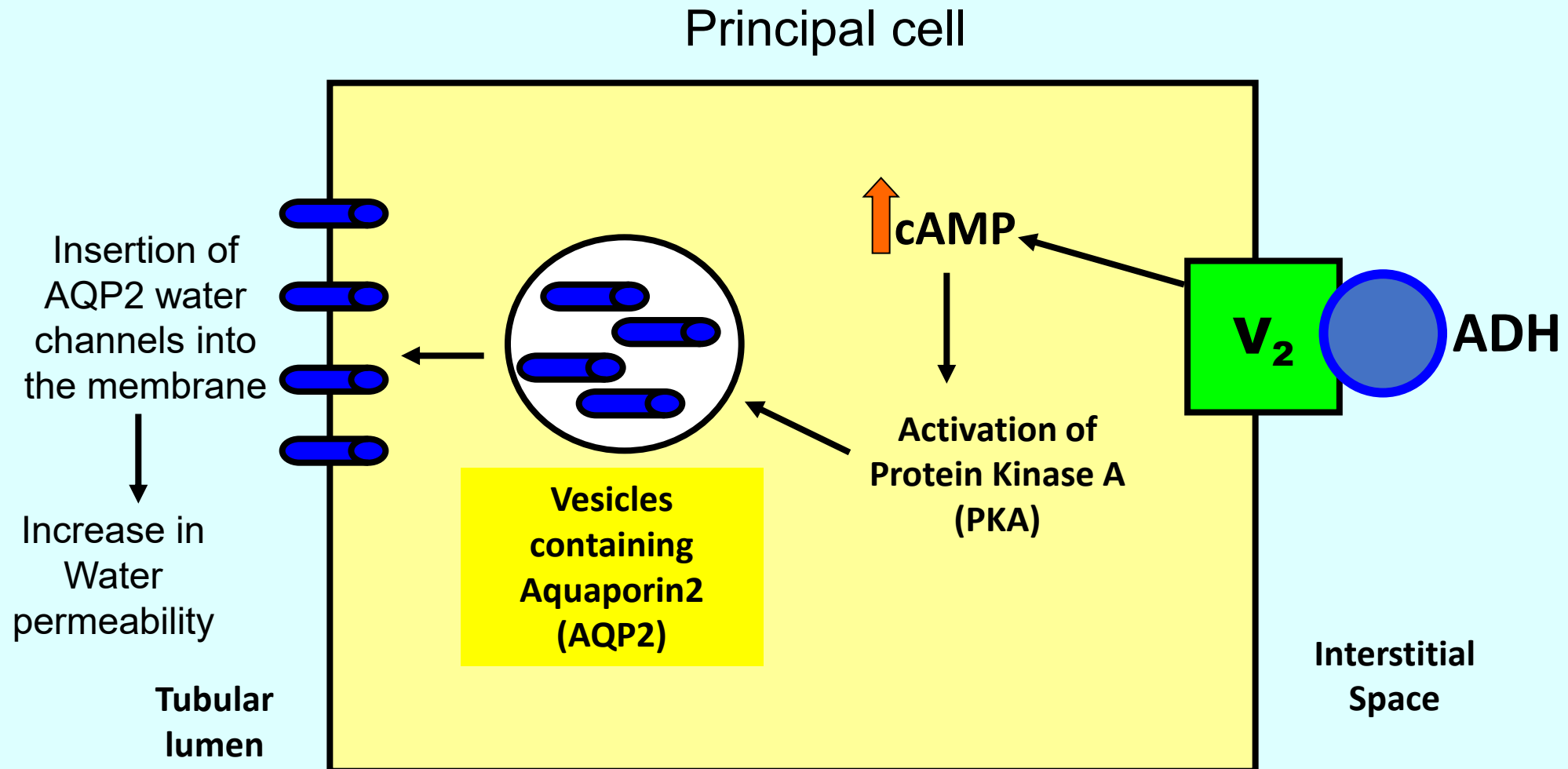
Decreased body fluid osmolality
Increased blood volume
Increased blood pressure
Atrial natriuretic peptide
Ethanol

Changes in ECF osmolality and ECF volume are the TWO major factors

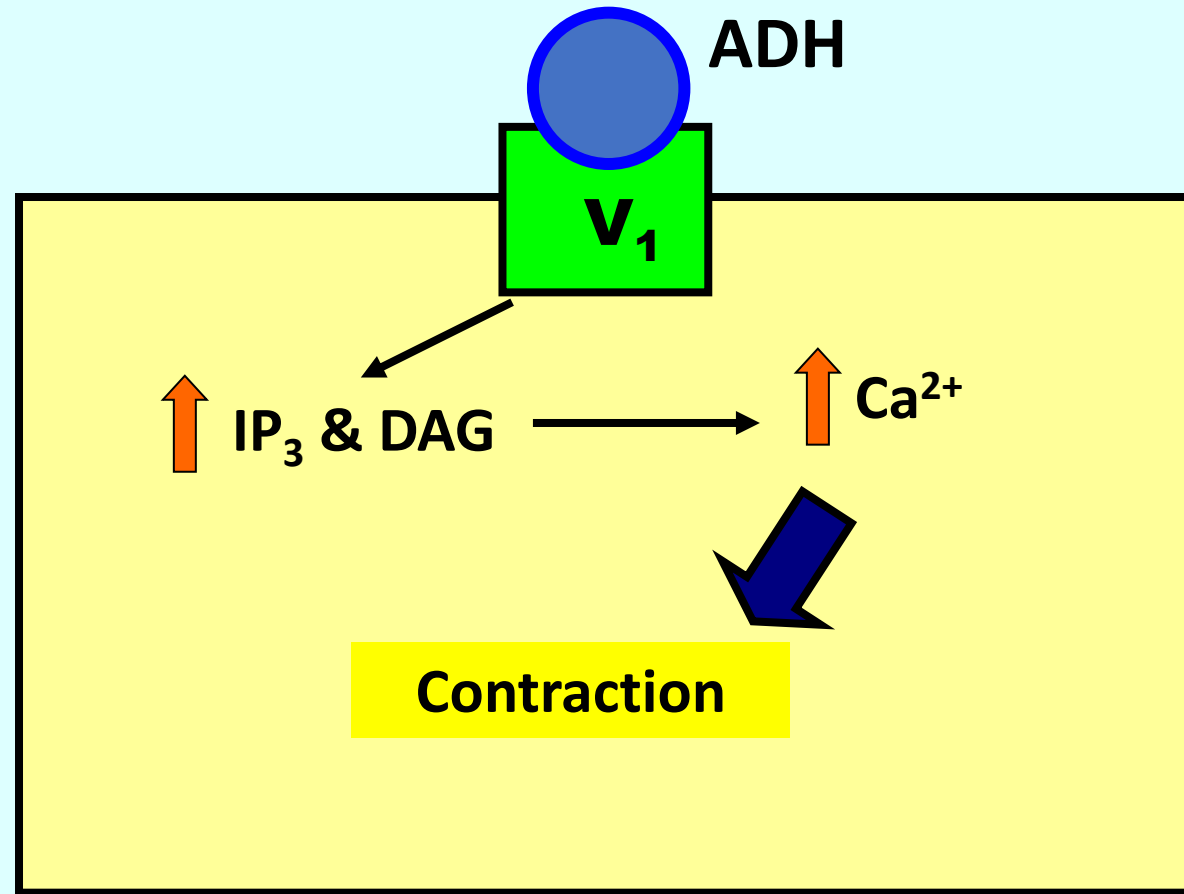
Factors regulating AVP (ADH) secretion



Mechanism of action of ADH in kidney (collecting duct): via V_2 receptors



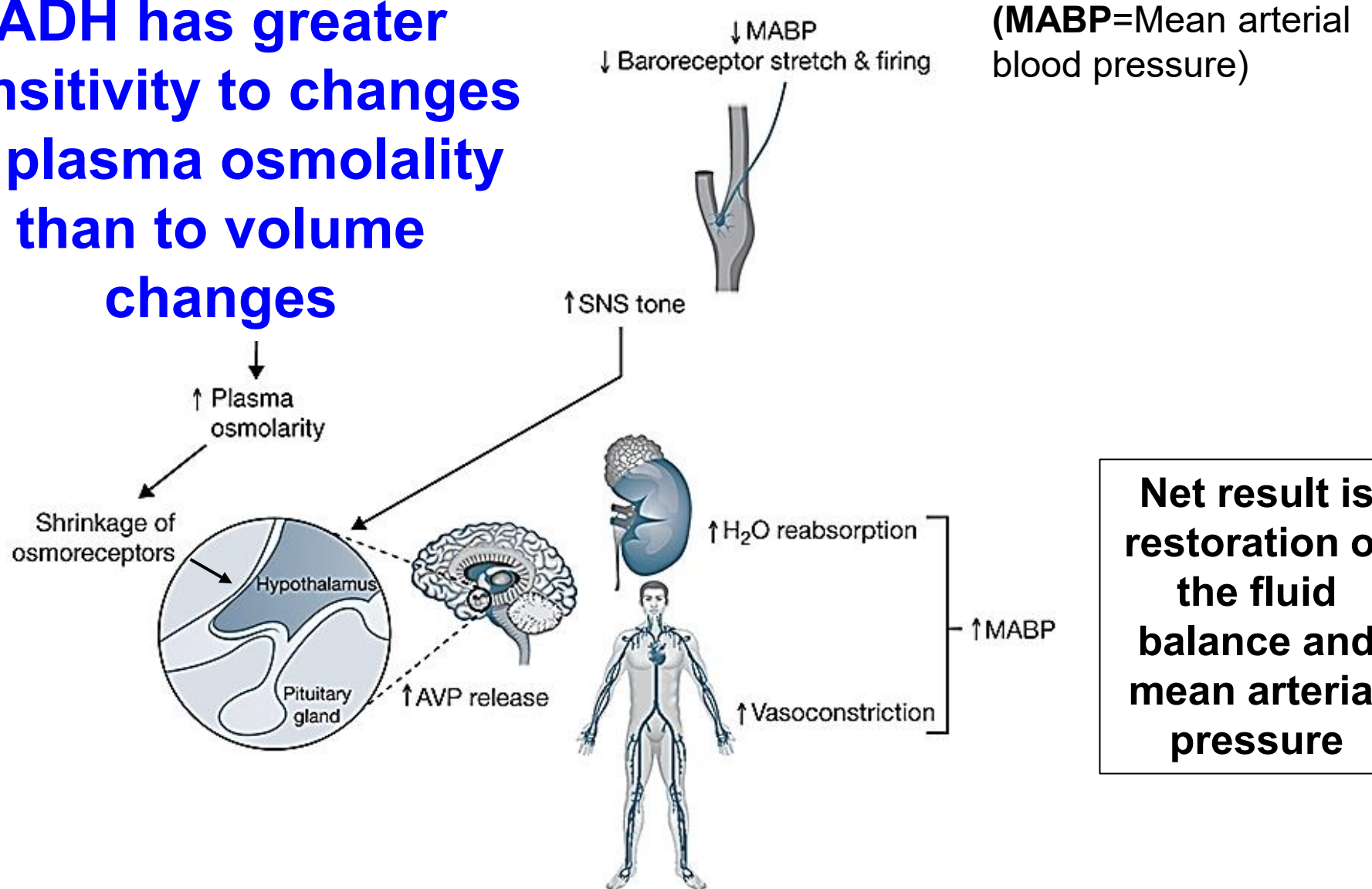
Mechanism of action of ADH in blood vessels (smooth muscles): via V_1 receptors



Vascular smooth muscle cell

ADH restores plasma osmolality and volume

ADH has greater sensitivity to changes in plasma osmolality than to volume changes



Summary of Physiologic effects of Vasopressin (ADH)

On Kidney:

- Increases water permeability of the DCT and collecting ducts,
- Increases urea permeability of the inner medullary collecting ducts,
- Increases activity of the $\text{Na}^+ - \text{K}^+ - 2\text{Cl}^-$ cotransporter.
- Effects, 2 and 3 enhance the osmotic gradient of the medullary interstitium in conditions where circulating levels of ADH are high (e.g., dehydration). This favors effect 1 which leads to water reabsorption in the distal nephron.

On Blood vessels:

- Produces contraction of the vascular smooth muscle cells and increases peripheral resistance.

Disorders associated with ADH

Decreased ADH action



Diabetes Insipidus (DI)



2 Types: Neurogenic
(Central) DI and
Nephrogenic DI

Increased ADH secretion



Syndrome of inappropriate
ADH (SIADH)

Diabetes Insipidus: Neurogenic versus Nephrogenic

In all patients with DI, **ADH action on kidneys is lacking** and therefore, patients present with:

Passing large volume of urine (polyuria) with low osmolarity (hypotonic urine), increased fluid intake (polydipsia), thirst, elevated plasma sodium and elevated plasma osmolarity.

Neurogenic or Central DI

- ADH secreting neurons are lost
- **↓ plasma ADH level**
- Mostly because of head injury and brain tumors
- Treated with ADH injection.

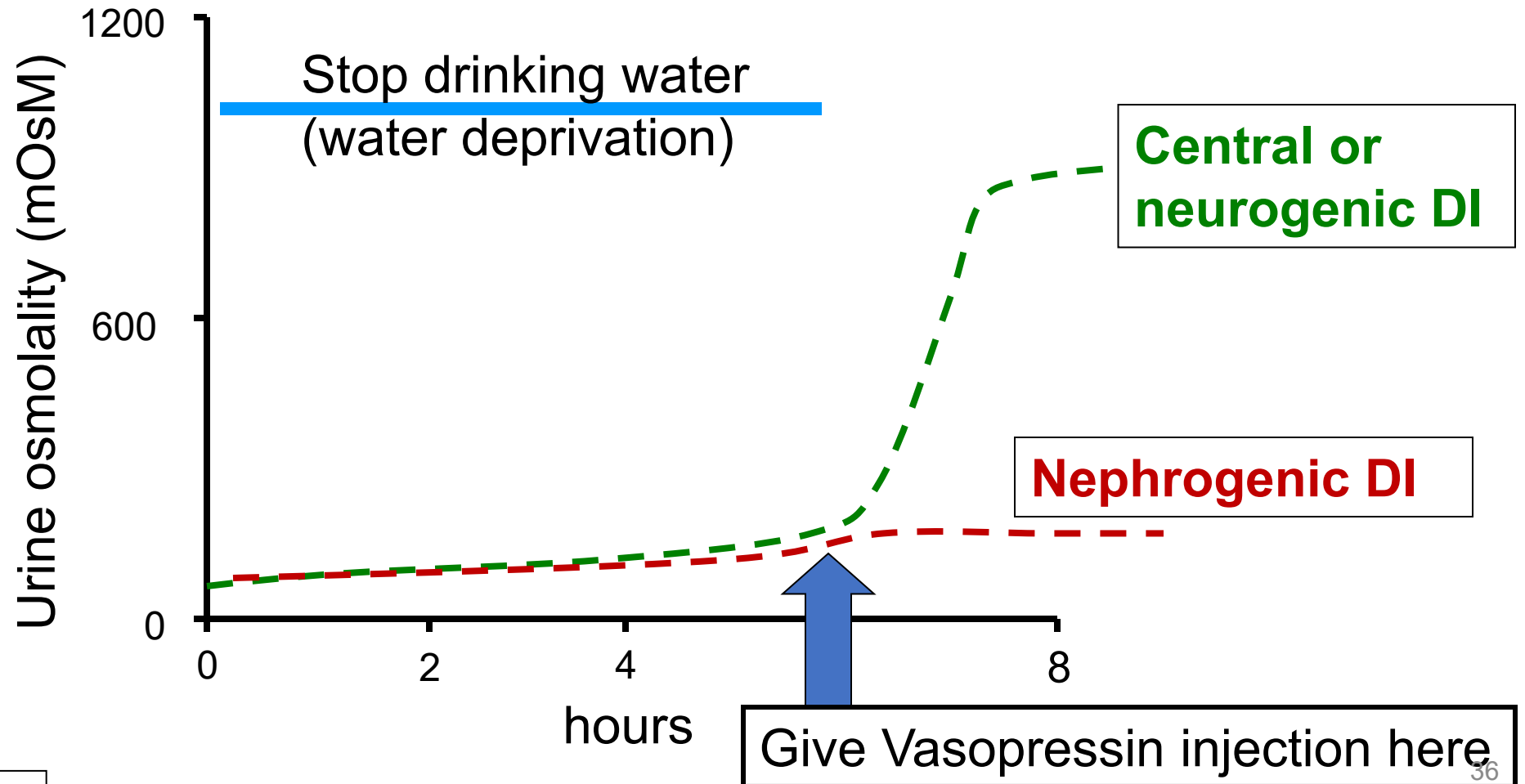
Nephrogenic DI

- Kidney is unable to respond to ADH
- **↑ Plasma ADH level**
- Mostly because of defective V2 receptors or drug-induced
- Treatment with ADH injection does not help.

Neurogenic (Central) vs Nephrogenic Diabetes Insipidus

How would you distinguish central DI from nephrogenic DI?

All patients have $\uparrow$ serum osmolality, $\uparrow$ serum sodium, $\uparrow$ urine volume, and hypotonic urine.



Syndrome of Inappropriate ADH or SIADH

- **Excess secretion of ADH** and its effect on the kidney
- Most common causes are head injury and lung cancer
- Plasma ADH is abnormally high
- Physiologic ADH regulation is not working
- Highly concentrated (hypertonic) urine with low volume
- **Low serum sodium and osmolality.**

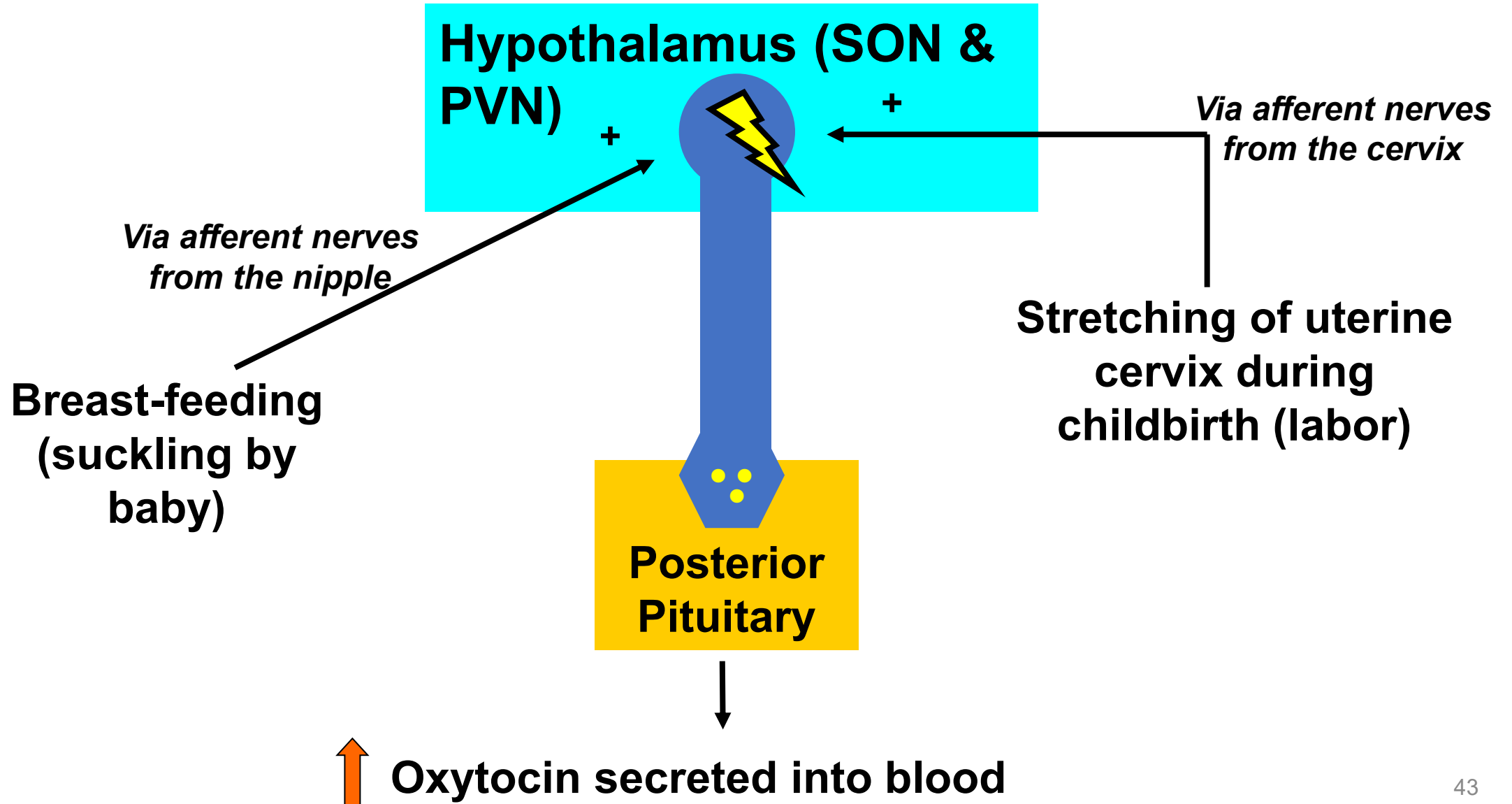
These patients have **normal BP** but
low serum sodium and osmolality!



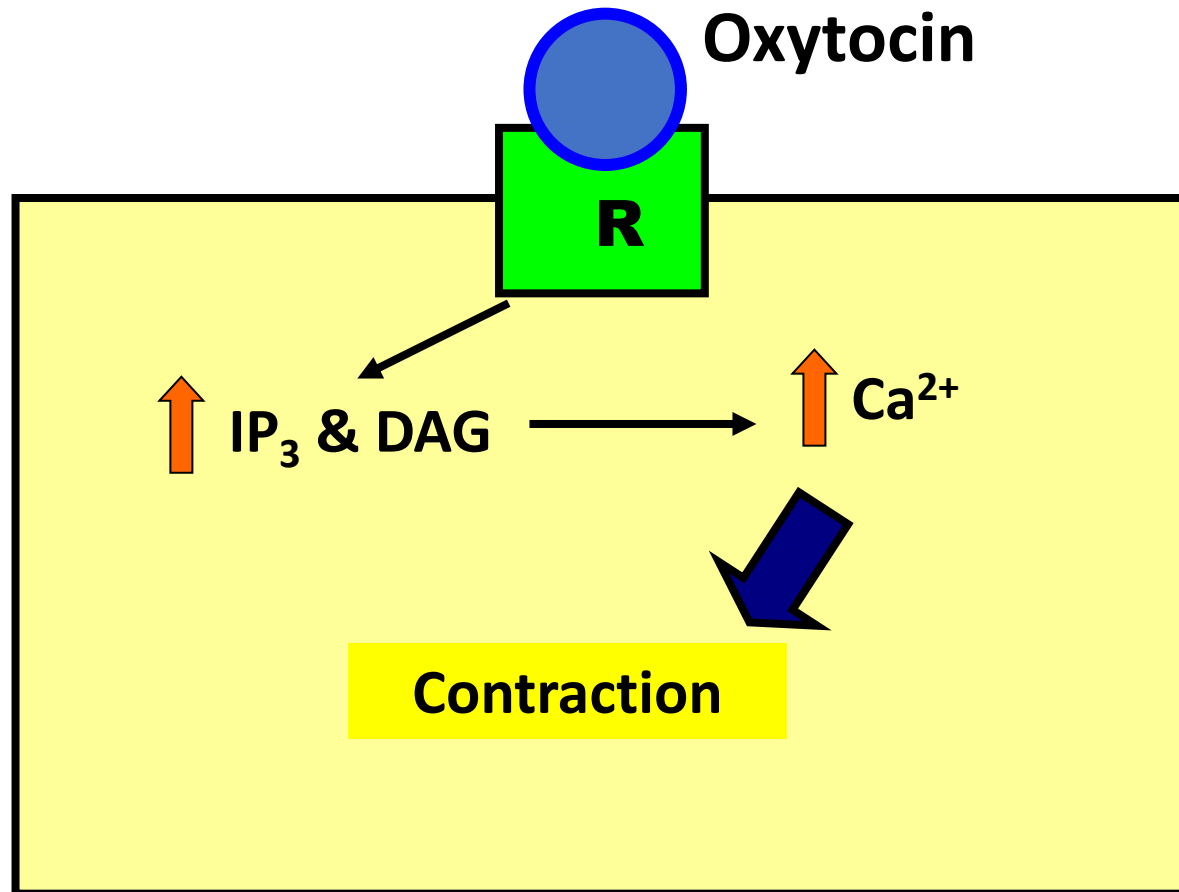
Lecture Outline

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- ❖ **Posterior pituitary hormone: Oxytocin**

Factors that cause Oxytocin secretion



Oxytocin: Target tissues and Mechanism of action



2 target tissues:

1. **Myoepithelial cells** of the mammary gland

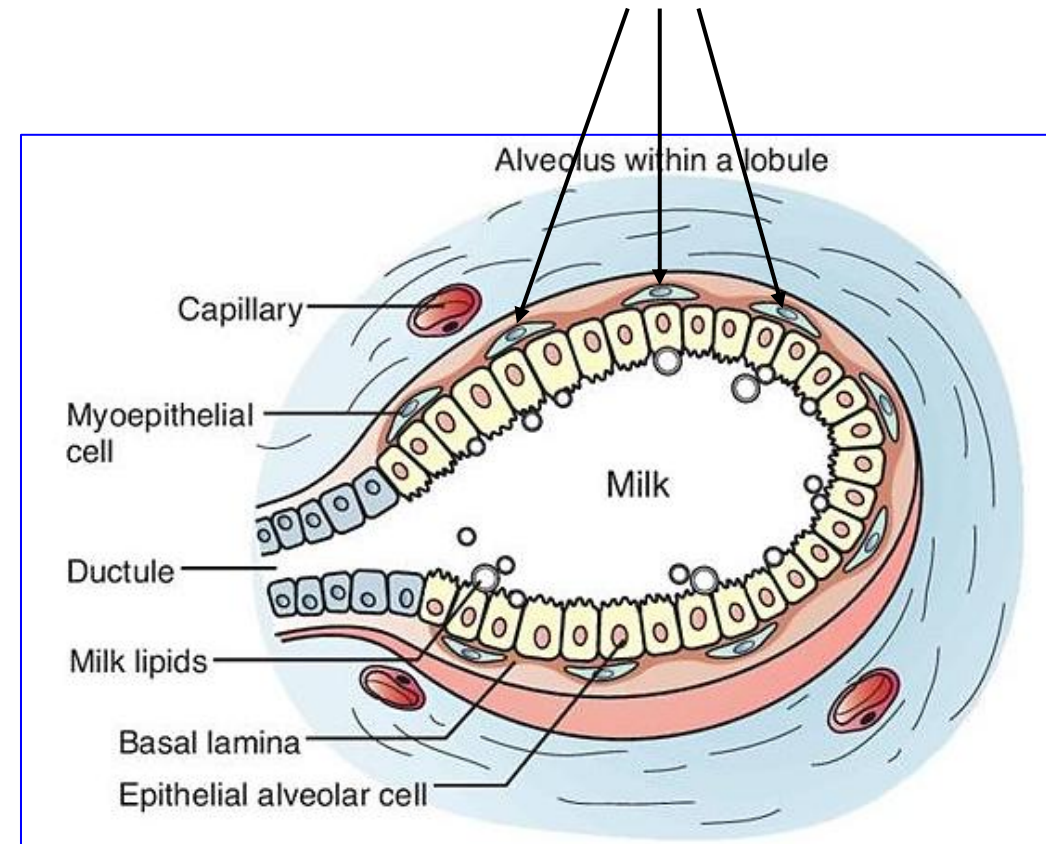
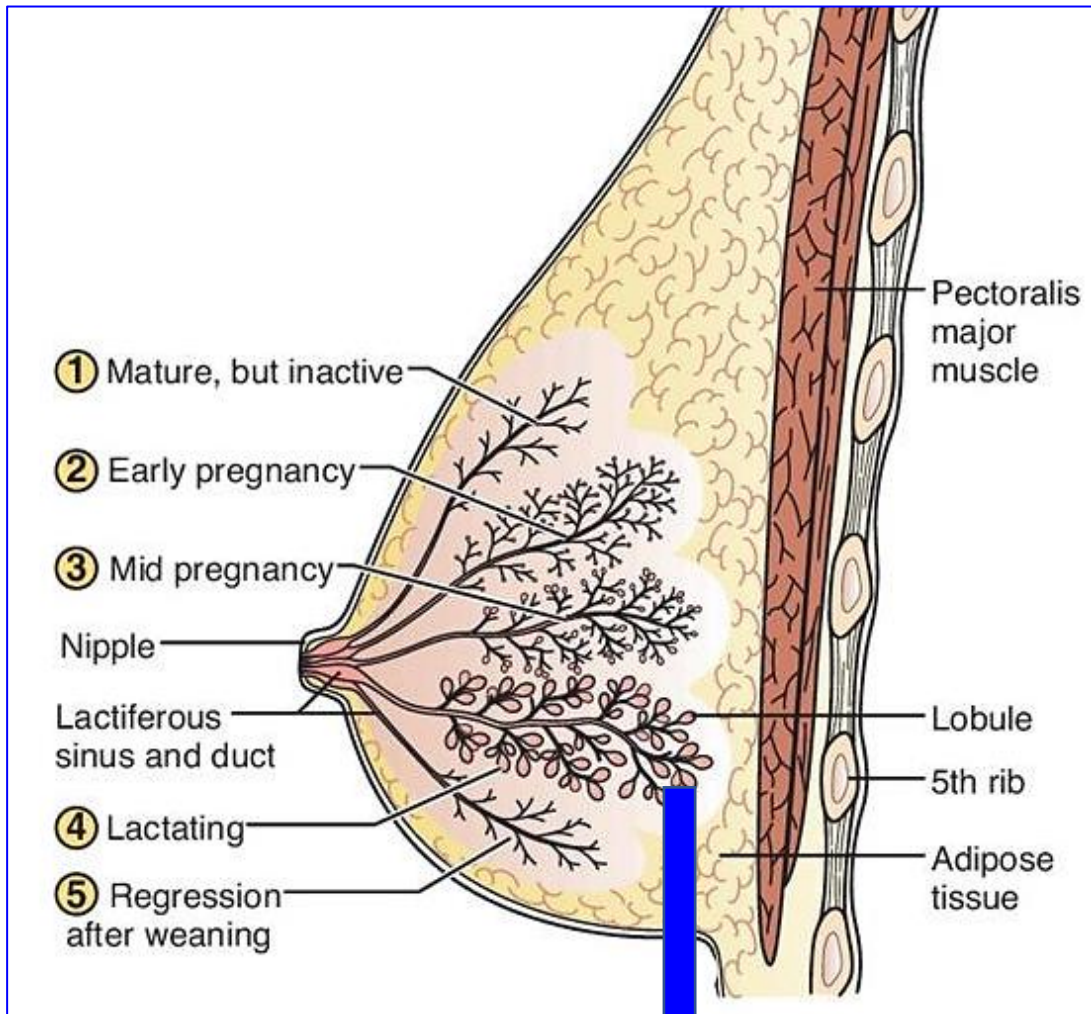
↓
Milk ejection

2. Smooth muscles of the uterus (**myometrium**)

↓
Expulsion of fetus and placenta.

Action of Oxytocin: Milk Ejection

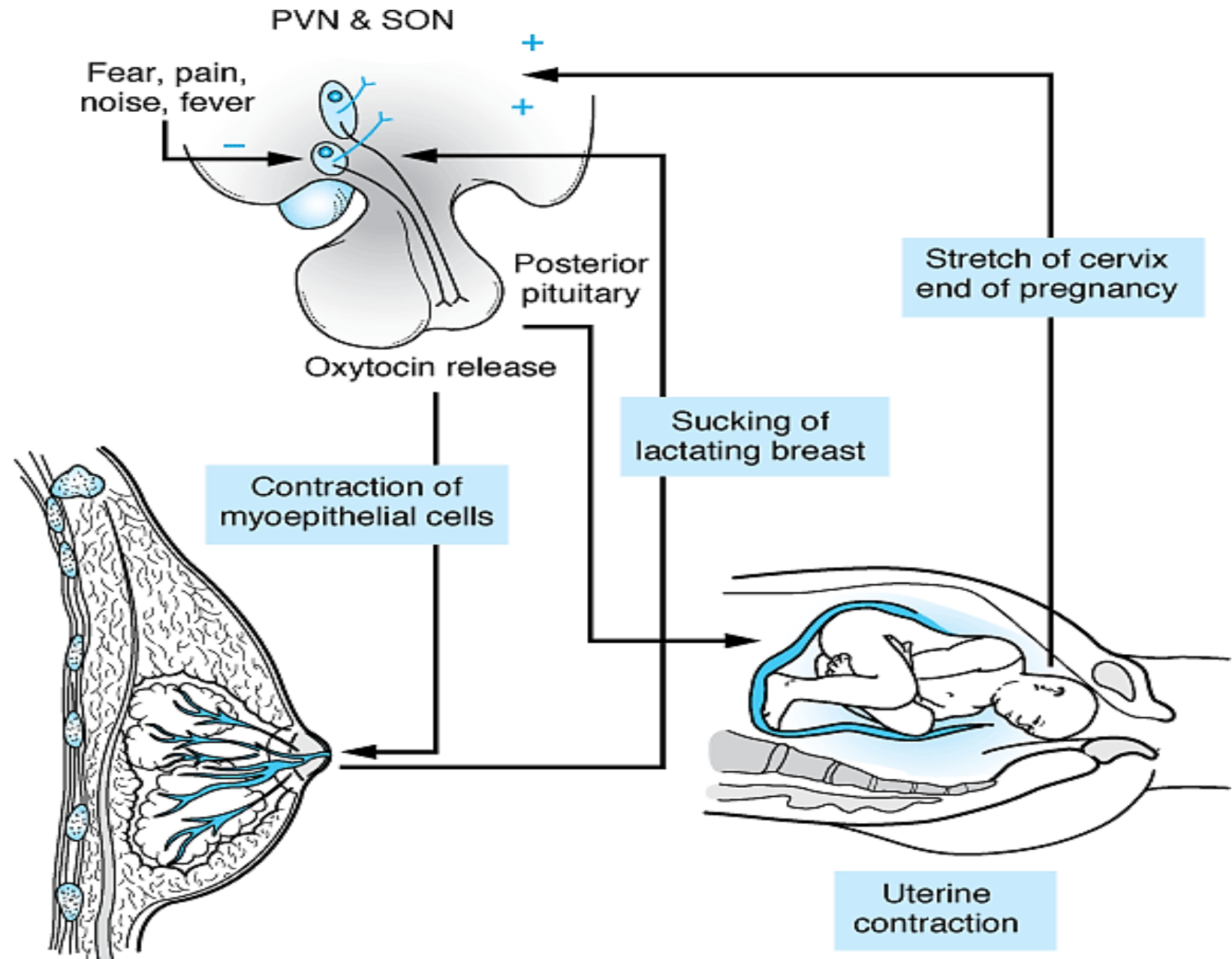
Contraction of the **Myoepithelial cells** surrounding the alveoli squeeze milk out of the mammary gland

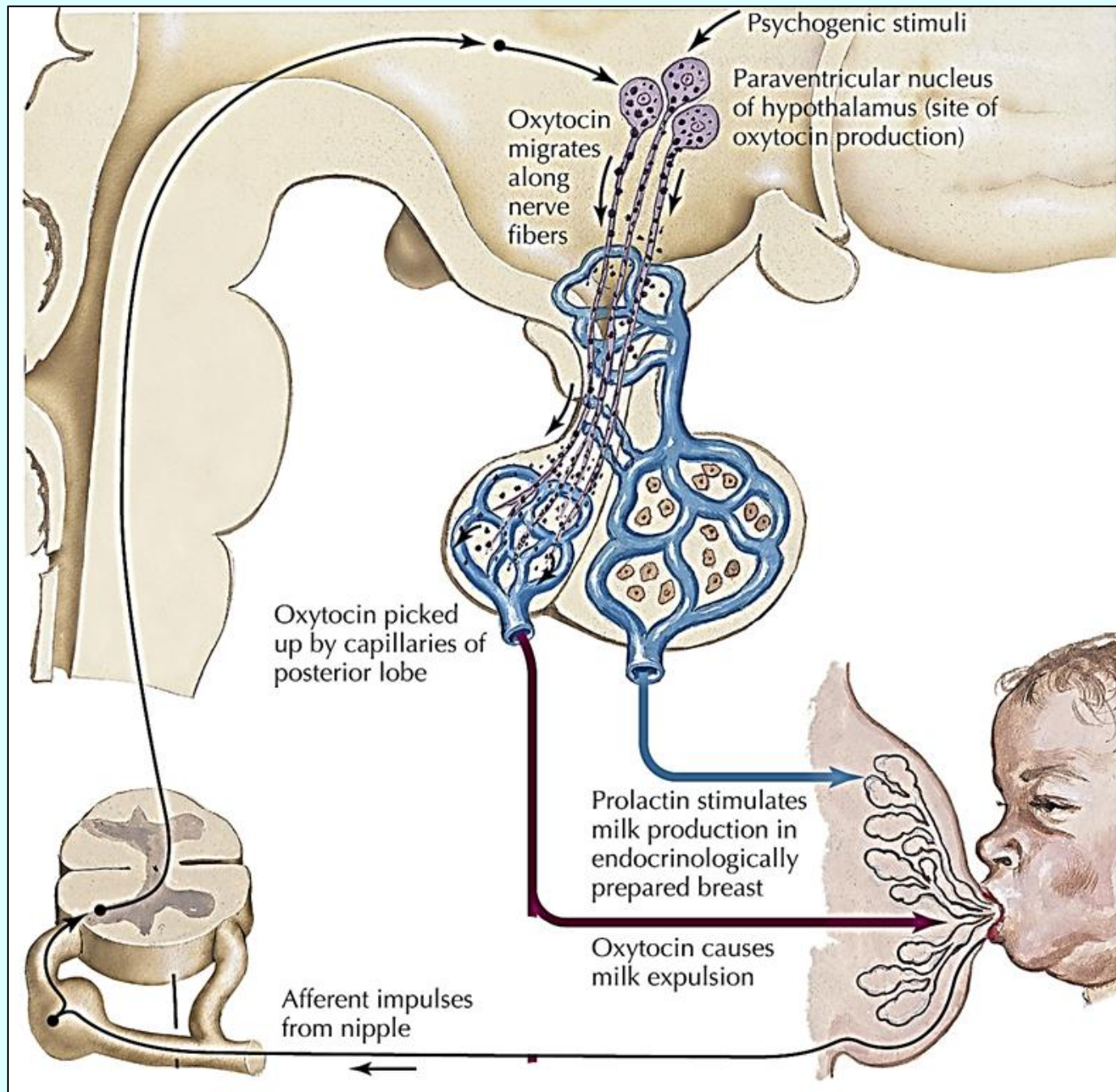


Oxytocin actions are related to childbirth and breast-feeding

Oxytocin secretion sets a positive feedback

Milk ejection is an example for a **Neuroendocrine Reflex**

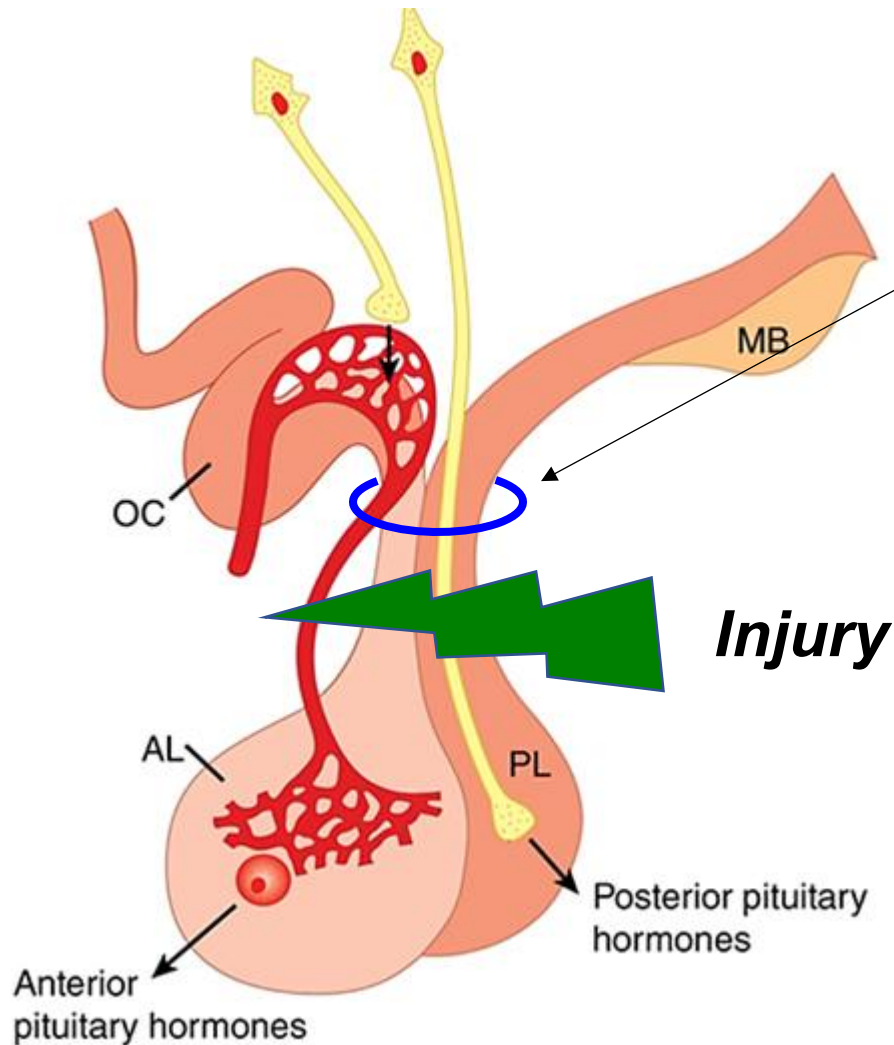




**Oxytocin release
by suckling
reflex**

**Stimulation of
sensory nerves in
the nipple by the
infant initiates the
suckling reflex**

Pituitary Stalk: Clinical importance



The pituitary stalk contains portal vessels and axons of the hypothalamic nuclei terminating in the posterior pituitary lobe.

Injury or compression (by a tumor or skull fracture) of the stalk can lead to:

- **Hypopituitarism,**
- **Diabetes insipidus, and**
- **Hyperprolactinemia.**

Other Recommended Readings

1. Section III: Endocrine & Reproductive Physiology. Ganong's Review of Medical Physiology, 26e. Kim E. Barrett, Susan M. Barman, Heddwen L. Brooks, Jason X.-J. Yuan.
2. General Principles of Endocrinology and Pituitary and Hypothalamic Hormones: Netter's Essential Physiology, Second Edition. Mulroney, Susan E., PhD.
3. Part 12: Endocrinology and Metabolism-Section 1. Endocrinology. Harrison's Principles of Internal Medicine, 21e. Joseph Loscalzo, Anthony Fauci, Dennis Kasper, Stephen Hauser, Dan Longo, J. Larry Jameson.

Any Questions ?

How to reach me: Either walk into my office or email.

Subramanya Upadhy, MD.

Professor,

**Depart of Physiology, Neuroscience & Behavioral
Sciences, Lower Charter Hall,
School of Medicine, St. George's University.**

Email: supadhyasgu@sgu.edu